THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 3

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The instrument that makes very small objects appear larger is the: 1

(a) telescope **(b)** microscope **(c)** periscope **(d)** stethoscope

2. Which of these is an example of social health? 1

(a) having a strong heart
(b) getting along well with family and neighbours
(c) having a normal body temperature
(d) being free from cough

3. The symbol of a switch in the open (OFF) position shows: 1

(a) a closed loop **(b)** a gap in the circuit
(c) a bulb **(d)** a cell

4. The SI unit of force is the: 1

(a) watt **(b)** pascal **(c)** newton **(d)** joule

5. The SI unit of pressure is the: 1

(a) joule **(b)** watt **(c)** newton **(d)** pascal

6. There are spaces between the particles of matter. Which observation supports this? 1

(a) ice floats
(b) metals shine
(c) wood burns
(d) sugar dissolves in water without raising the water level much

7. Which of the following is an element? 1

(a) H_2O **(b)** CO_2 **(c)** O_2 **(d)** $NaCl$

8. In a solution of sugar in water, sugar is the: 1

(a) precipitate **(b)** solvent **(c)** solution **(d)** solute

9. According to the second law of reflection: 1

(a) angle of reflection is zero
(b) angle of incidence equals angle of reflection
(c) angles are always 90 degrees
(d) angle of incidence is double the angle of reflection

10. The Earth completes one rotation on its axis in about: 1

(a) 24 hours **(b)** 365 days **(c)** 12 hours **(d)** 27 days

11. Which of the following is an abiotic component? 1

(a) sunlight **(b)** grass **(c)** fungi **(d)** earthworm

12. The outermost layer of the Earth is the: 1

(a) inner core **(b)** crust **(c)** outer core **(d)** mantle

13. The scientific method is best described as: 1

(a) copying a friend's result
(b) a fixed list of steps always followed in one order
(c) a flexible approach that may be repeated and revised
(d) guessing the answer

14. A group of similar cells performing a particular function is a: 1

(a) organism **(b)** organ **(c)** organelle **(d)** tissue

15. A disease that lasts for a long time is called: 1

(a) acute **(b)** chronic **(c)** viral **(d)** sudden

16. A fuse wire has: 1

(a) no resistance **(b)** low melting point
(c) magnetic property **(d)** high melting point

17. Which of these reduces friction? 1

(a) dust **(b)** sand **(c)** lubricants **(d)** rough paper

18. Moving air is called: 1

(a) cloud **(b)** fog **(c)** wind **(d)** rain

19. The change from liquid to gas at any temperature below the boiling point is: 1

(a) freezing **(b)** melting **(c)** sublimation **(d)** evaporation

20. The property of metals by which they can be drawn into wires is: 1

(a) ductility **(b)** solubility **(c)** brittleness **(d)** malleability

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. List the three main layers of the Earth. 2

22. Why do scientists repeat an experiment several times? 2

23. Why is the nucleus called the control centre of the cell? 2

24. Why are vaccines given to children? 2

25. Why is the filament of a bulb made of tungsten? 2

26. Why do we sprinkle powder on a carrom board? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. Explain how a fuse protects a circuit. 3

28. Describe how an electroscope works. 3

29. Why does a balloon expand when more air is blown in? 3

30. Explain the cooling effect of evaporation using particles. 3

31. Explain why water is a compound and not a mixture of hydrogen and oxygen. 3

OR

How much solute is needed to prepare 200 g of a 10 per cent salt solution? 3

32. Differentiate between a concave and a convex mirror in three points. 3

OR

Explain how the seasons are caused. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. Explain the working of a barometer and the variation of atmospheric pressure with altitude and weather. 5

34. Describe an experiment to study the cooling by evaporation using cotton wool dipped in different liquids on a thermometer bulb. 5

35. Explain atoms and molecules with the help of H_2, O_2, H_2O and CO_2, giving the number of atoms in each. 5

36. Describe the types of solutions on the basis of the physical states of solute and solvent with an example each. 5

OR

Describe the lens types, their action on light, the names of the principal focus and the focal length with uses. 5

37. Explain why we see the Moon from Earth in phases and not always the full disc, using the positions of the Sun, Earth and Moon. 5

OR

Explain how the removal of one species affects an ecosystem. Give an example of vultures or tigers. 5

38. Explain the importance of the ozone layer, the harmful gases and the Montreal Protocol. 5

OR

Explain the steps of the scientific method using the puri-puffing investigation. Why is it called a cycle? 5

END OF QUESTION PAPER Total: 80 Marks

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